

Radar HRRP Target Recognition Based on Linear Dynamic Model

Wang Penghui, Du Lan*, Pan Mian, Zhang Xuefeng and Liu Hongwei

National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China

{wangpenghui, dulan}@mail.xidian.edu.cn

Abstract

A novel recognition method is proposed to relax the heavy requirement of training set size in the radar High Resolution Range Profile (HRRP) target recognition. Firstly, the statistical characteristic of HRRP's frequency spectrum is analyzed and considered to be a wide-sense stationary process. Then a linear dynamic model is introduced to describe the frequency spectrum amplitude and an Expectation Maximization (EM) algorithm is employed to estimate model parameters. Finally, experimental results show the proposed method can obtain satisfactory recognition accuracy and rejection performance with only a few training samples.

Keywords: Radar target recognition; High Resolution Range Profile (HRRP); Wide Sense Stationary-Unrelated Scattering (WSS-US); Linear Dynamic Model (LDM).

1. Introduction

Radar automatic target recognition (RATR) is to identify the unknown target from its radar echoes. A high-resolution range profile (HRRP) is the coherent summations of the complex time returns from target scatterers in each range cell. Since it contains abundant target structure information, radar HRRP target recognition has received intensive attention from the RATR community [1-8].

For broadband radar, the target signal usually occupies lots of range cells in the return and thus the HRRPs of the target are high-dimensional, which leads to the curse of dimensionality [9] in radar HRRP recognition. To overcome this problem, usually lots of HRRP training samples are needed. However, for non-cooperative targets, it is expensive or even impractical to gather sufficient HRRPs. So how to obtain acceptable recognition performance with a small training set is an urgent task to be tackled.

Many relevant researches have been carried out, which mainly focus on the following two aspects: 1. Employing statistical models with low degrees of freedom. For example, Ref. [1] used the Adaptive Gaussian Classifier (AGC) model to describe HRRPs. In Ref. [2], HRRPs was modeled by the Factor Analysis (FA) model. Although showing promising recognition performance, both of these models cannot get rid of curse of dimensionality thoroughly; 2. Extracting low-dimensional features. Ref. [3] projected HRRPs to a low-dimensional space for recognition. Ref. [4] computed central moments of HRRPs as discriminating feature. Even though these features significantly relax the requirement of training set size, some discriminating information is unavoidably lost in the dimension reduction process and the recognition performance deteriorates.

This paper proposes to use a linear dynamic model to characterize the frequency spectrum amplitude of HRRP with the model parameters estimated by Expectation Maximization algorithm. All the parameters can be estimated with only one HRRP, which significantly alleviate the requirement of training sample size.

2. Statistical Characterization of HRRP

An HRRP $\mathbf{x}$ is the coherent summations of the complex returns from target scatterers in each range cell, which can be represented as:

$$\mathbf{x} = e^{j\varphi} [x_1, x_2, \dots, x_d]^T \text{ with } x_n = \sum_{i=1}^{I_n} \alpha_{in} e^{j\phi_{in}}, \quad n=1, 2, \dots, d \quad (1)$$

where $[\cdot]^T$ denotes matrix transpose, φ is the initial phase, d is the number of range cells, i.e. the dimensionality of HRRP, I_n is the total number of scatterers in the n th range cell, and α_{in} and ϕ_{in} are the reflectivity and phase of the i th scatterer in the n th range cell. The frequency spectrum $\mathbf{y}$ of $\mathbf{x}$ is defined as:

$$\mathbf{y} = \text{fft}(\mathbf{x}) = e^{j\varphi} [y_1, y_2, \dots, y_d]^T$$
$$\text{with } y_f = \sum_{n=1}^d x_n e^{-j2\pi f n/d}, \quad f=1, 2, \dots, d \quad (2)$$

where $\text{fft}(\cdot)$ denotes the fast Fourier transform (FFT) and y_f is the f th spectral component. The feasibility of frequency spectrum based recognition has been investigated in Refs. [5, 6]. In this paper we will adopt frequency spectrum of HRRP for recognition due to its following advantages: 1. There is no information lost in the transformation process from HRRP in time domain to the frequency spectrum in frequency domain; 2. Frequency spectrum amplitude is translation invariant, so translation alignment is not needed in the recognition process which simplifies the recognition system; 3. HRRPs represented in the frequency domain may exhibit properties which cannot be seen in the time domain (such as the stationarity introduced subsequently).

According to the Wide Sense Stationary-Unrelated Scattering (WSS-US) model [1, 6, 10], the return in each range cell can be regarded as independent zero-mean complex Gaussian variables. After some mathematical derivation, we can get the conclusion that correlation function of two spectral components only depends on their frequency difference, i.e., the frequency spectrum is a wide sense stationary (WSS) process. The differences in target structure will lead to distinct WSS processes embedded in frequency spectra, which can be used for recognition. To get rid of the initial phase sensitivity [2], we discard the phase information

and only consider the spectral amplitude $\mathbf{z} = |\mathbf{y}|$. It is simple to prove that $\mathbf{z}$ is still a WSS process.

3. Linear Dynamic Model

Linear Dynamic Model (LDM), also called state space model, is widely used in WSS process modeling and has been successfully applied in recognition area [11]. Given the spectral amplitude $\mathbf{z}$ of an HRRP, the LDM is consisted of the following observation equation and state equation:

$$\begin{aligned} z_f &= \mathbf{C}\mathbf{u}_f + \mu + v_f \\ \mathbf{u}_f &= \mathbf{A}\mathbf{u}_{f-1} + \mathbf{w}_f \end{aligned} \quad (3)$$

where z_f is the f th element in $\mathbf{z}$, $f=1,2,\dots,d$, $\mathbf{C}$ is the $1 \times m$ dimensional observation matrix ($m \ll d$), $m \times 1$ dimensional vector $\mathbf{u}_f$ is the hidden state of z_f , μ is the mean of $\mathbf{z}$, $v_f \sim G(v_f | 0, \sigma^2)$ is the observation noise, $v_f \sim G(v_f | 0, \sigma^2)$ denotes v_f is Gaussian distributed with mean 0 and variance σ^2 , $\mathbf{A}$ is the $m \times m$ dimensional state transition matrix, $m \times 1$ dimensional $\mathbf{w}_f \sim G(\mathbf{w}_f | \mathbf{0}, \mathbf{Q})$ is state noise and is uncorrelated with v_f , $\mathbf{u}_0 \sim G(\mathbf{u}_0 | \boldsymbol{\mu}_0, \mathbf{R})$ is initial state. To reduce the degree of freedom of LDM, we constrain $\mathbf{Q}$ and $\mathbf{R}$ to be diagonal matrices.

From Eq.(3) we can see the frequency evolution information of the spectral amplitude (i.e., wide-sense stationarity) is embedded in the hidden Markov state sequence $\mathbf{U} = \{\mathbf{u}_0, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_d\}$. Compared with those models who describe the wide-sense stationarity directly (e.g., autoregressive model), LDM has better recognition performance under noisy conditions [11].

All the model parameters $\boldsymbol{\Theta} = \{\mathbf{A}, \mathbf{C}, \mathbf{Q}, \mathbf{R}, \boldsymbol{\mu}_0, \sigma^2, \mu\}$ can be estimated by Expectation Maximization (EM) algorithm. The EM algorithm is very stable and can converge quickly. It iteratively maximizes the expected joint log-likelihood of spectral amplitude $\mathbf{z}$ and hidden state sequence $\mathbf{U}$ to obtain the parameters estimation $\tilde{\boldsymbol{\Theta}} = \{\tilde{\mathbf{A}}, \tilde{\mathbf{C}}, \tilde{\mathbf{Q}}, \tilde{\mathbf{R}}, \tilde{\boldsymbol{\mu}}_0, \tilde{\sigma}^2, \tilde{\mu}\}$. The expected joint log-likelihood $Q(\boldsymbol{\Theta})$ is defined as:

$$\begin{aligned} Q(\boldsymbol{\Theta}) &= E(\ln p(\mathbf{z}, \mathbf{U}) | \mathbf{z}) \\ &= -\frac{1}{2} \sum_{f=1}^d \text{Tr}[E(\mathbf{u}_f \mathbf{u}_f^T | \mathbf{z}) \mathbf{Q}^{-1} + E(\mathbf{u}_{f-1} \mathbf{u}_{f-1}^T | \mathbf{z}) \mathbf{A}^T \mathbf{Q}^{-1} \mathbf{A} - 2E(\mathbf{u}_{f-1} \mathbf{u}_f^T | \mathbf{z}) \mathbf{Q}^{-1} \mathbf{A}] \\ &\quad - \frac{1}{2\sigma^2} \sum_{f=1}^d \{\text{Tr}[E(\mathbf{u}_f \mathbf{u}_f^T | \mathbf{z}) \mathbf{C}^T \mathbf{C} - 2E(\mathbf{u}_f^T | \mathbf{z}) \mathbf{C}^T (z_f - \mu)] + (z_f - \mu)^2\} \\ &\quad - \frac{1}{2} \{\text{Tr}[E(\mathbf{u}_0 \mathbf{u}_0^T | \mathbf{z}) \mathbf{R}^{-1} - 2\boldsymbol{\mu}_0^T E(\mathbf{u}_0^T | \mathbf{z}) \mathbf{R}^{-1}] + \boldsymbol{\mu}_0^T \mathbf{R}^{-1} \boldsymbol{\mu}_0\} - \frac{1}{2} \ln |\mathbf{R}| \\ &\quad - \frac{d(m+1)+1}{2} \ln(2\pi) - \frac{d}{2} \ln |\mathbf{Q}| - \frac{d}{2} \ln |\sigma^2| \end{aligned} \quad (4)$$

where $E(\cdot | \cdot)$ denotes the conditional expectation operator and $\text{Tr}(\cdot)$ is the matrix trace operator. The detailed implementation of EM algorithm is given in Algorithm 1:

Algorithm 1: EM algorithm

Step1. Let iteration index $\tau=1$, set convergence threshold ε ($0 < \varepsilon < 1$) and randomly initialize model parameters $\tilde{\boldsymbol{\Theta}}(1) = \{\tilde{\mathbf{A}}(1), \tilde{\mathbf{C}}(1), \tilde{\mathbf{Q}}(1), \tilde{\mathbf{R}}(1), \tilde{\boldsymbol{\mu}}_0(1), \tilde{\sigma}^2(1), \tilde{\mu}(1)\}$.

Step2. E-step: Given the parameters $\tilde{\boldsymbol{\Theta}}(\tau) = \{\tilde{\mathbf{A}}(\tau), \tilde{\mathbf{C}}(\tau), \tilde{\mathbf{Q}}(\tau), \tilde{\mathbf{R}}(\tau), \tilde{\boldsymbol{\mu}}_0(\tau), \tilde{\sigma}^2(\tau), \tilde{\mu}(\tau)\}$ estimated in the τ th iteration, use Kalman forward filtering and RTS backward smoother to estimate $E(\mathbf{u}_f^T | \mathbf{z})$, $E(\mathbf{u}_f \mathbf{u}_f^T | \mathbf{z})$ and $E(\mathbf{u}_{f-1} \mathbf{u}_f^T | \mathbf{z})$.

Step2.1 Given spectral amplitude sequence $\tilde{\mathbf{z}}_1^k = \{z_1, z_2, \dots, z_k\}$ ($0 < k \leq d$, $\mathbf{z} = \tilde{\mathbf{z}}_1^d$), define the following notations:

$$\begin{aligned} \mathbf{u}_{f|k} &= E(\mathbf{u}_f | \tilde{\mathbf{z}}_1^k); \\ \boldsymbol{\Sigma}_{f|k} &= E[(\mathbf{u}_f - \mathbf{u}_{f|k})(\mathbf{u}_f - \mathbf{u}_{f|k})^T | \tilde{\mathbf{z}}_1^k]; \\ \boldsymbol{\Sigma}_{f,f-1|k} &= E[(\mathbf{u}_f - \mathbf{u}_{f|k})(\mathbf{u}_{f-1} - \mathbf{u}_{f-1|k})^T | \tilde{\mathbf{z}}_1^k]; \\ \mathbf{P}_{f|k} &= E[\mathbf{u}_f^T \mathbf{u}_f | \tilde{\mathbf{z}}_1^k]; \quad \mathbf{P}_{f,f-1|k} = E[\mathbf{u}_f^T \mathbf{u}_{f-1} | \tilde{\mathbf{z}}_1^k]. \end{aligned}$$

Using Kalman forward filtering, we have

$$\begin{aligned} \mathbf{u}_{f|f-1} &= \tilde{\mathbf{A}}(\tau) \mathbf{u}_{f-1|f-1} \\ \boldsymbol{\Sigma}_{f|f-1} &= \tilde{\mathbf{A}}(\tau) \boldsymbol{\Sigma}_{f-1|f-1} \tilde{\mathbf{A}}(\tau)^T + \tilde{\mathbf{Q}}(\tau) \\ \mathbf{K}_f &= \boldsymbol{\Sigma}_{f|f-1} \tilde{\mathbf{C}}(\tau)^T [\tilde{\mathbf{C}}(\tau) \boldsymbol{\Sigma}_{f|f-1} \tilde{\mathbf{C}}(\tau)^T + \tilde{\sigma}^2(\tau)]^{-1} \\ \mathbf{u}_{f|f} &= \mathbf{u}_{f|f-1} + \mathbf{K}_f [z_f - \tilde{\mu}(\tau) - \tilde{\mathbf{C}}(\tau) \mathbf{u}_{f|f-1}] \\ \boldsymbol{\Sigma}_{f|f} &= \boldsymbol{\Sigma}_{f|f-1} - \mathbf{K}_f \tilde{\mathbf{C}}(\tau) \boldsymbol{\Sigma}_{f|f-1} \end{aligned} \quad (5)$$

where $\mathbf{u}_{0|0} = \tilde{\boldsymbol{\mu}}_0(\tau)$, $\boldsymbol{\Sigma}_{0|0} = \tilde{\mathbf{R}}(\tau)$, $\mathbf{I}_m$ is an $m \times m$ identity matrix.

Step2.2 Using RTS backward smoother, we have

$$\begin{aligned} \boldsymbol{\Sigma}_{d,d-1|d} &= [\mathbf{I}_m - \mathbf{K}_d \tilde{\mathbf{C}}(\tau)] \tilde{\mathbf{A}}(\tau) \boldsymbol{\Sigma}_{d-1|d-1} \\ \mathbf{J}_{f-1} &= \boldsymbol{\Sigma}_{f-1|f-1} \tilde{\mathbf{A}}(\tau)^T (\boldsymbol{\Sigma}_{f|f-1})^{-1} \\ \mathbf{u}_{f-1|d} &= \mathbf{u}_{f-1|f-1} + \mathbf{J}_{f-1} [\mathbf{u}_{f|d} - \tilde{\mathbf{A}}(\tau) \mathbf{u}_{f-1|f-1}] \\ \boldsymbol{\Sigma}_{f-1|d} &= \boldsymbol{\Sigma}_{f-1|f-1} + \mathbf{J}_{f-1} [\boldsymbol{\Sigma}_{f|d} - \boldsymbol{\Sigma}_{f|f-1}] \mathbf{J}_{f-1}^T \\ \boldsymbol{\Sigma}_{f-1,f-2|d} &= \boldsymbol{\Sigma}_{f-1|f-1} \mathbf{J}_{f-2}^T + \mathbf{J}_{f-1} [\boldsymbol{\Sigma}_{f,f-1|d} - \tilde{\mathbf{A}}(\tau) \mathbf{u}_{f-1|f-1}] \mathbf{J}_{f-2}^T \\ \mathbf{P}_{f|d} &= \boldsymbol{\Sigma}_{f|d} + \mathbf{u}_{f|d} (\mathbf{u}_{f|d})^T \\ \mathbf{P}_{f,f-1|d} &= \boldsymbol{\Sigma}_{f,f-1|d} + \mathbf{u}_{f|d} (\mathbf{u}_{f-1|d})^T \end{aligned} \quad (6)$$

Step3. M-step: Substituting $\mathbf{u}_{f|d}$, $\mathbf{P}_{f|d}$ and $\mathbf{P}_{f,f-1|d}$ in to Eq.(4), we can get the maximum likelihood estimation of all parameters

$$\begin{aligned} \tilde{\mu}(\tau+1) &= \frac{1}{d} \sum_{f=1}^d z_f \\ \tilde{\mathbf{A}}(\tau+1) &= \left(\sum_{f=1}^d \mathbf{P}_{f,f-1|d} \right) \left(\sum_{f=1}^d \mathbf{P}_{f-1|d} \right)^{-1} \\ \tilde{\mathbf{C}}(\tau+1) &= \left[\sum_{f=1}^d (z_f - \tilde{\mu}(\tau+1)) \mathbf{u}_{f|d}^T \right] \left(\sum_{f=1}^d \mathbf{P}_{f|d} \right)^{-1} \\ \tilde{\mathbf{Q}}(\tau+1) &= \frac{1}{d} \text{diag} \left(\sum_{f=1}^d \mathbf{P}_{f|d} - \tilde{\mathbf{A}}(\tau+1) \sum_{f=1}^d \mathbf{P}_{f,f-1|d}^T \right) \\ \tilde{\mathbf{R}}(\tau+1) &= \text{diag}(\mathbf{P}_{1|d} - \mathbf{u}_{0|d} \mathbf{u}_{0|d}^T) \\ \tilde{\sigma}^2(\tau+1) &= \frac{1}{d} \sum_{f=1}^d [(z_f - \tilde{\mu}(\tau+1))^2 - \tilde{\mathbf{C}}(\tau+1) \mathbf{u}_{f|d} (z_f - \tilde{\mu}(\tau+1))] \\ \tilde{\boldsymbol{\mu}}_0(\tau+1) &= \mathbf{u}_{0|d}, \end{aligned} \quad (7)$$

where $\text{diag}(\cdot)$ sets all off-diagonal elements of a matrix to zero.

Step3. Calculate $Q(\tilde{\theta}(\tau+1))$ by Eq.(4), if $\left| \frac{Q(\tilde{\theta}(\tau+1)) - Q(\tilde{\theta}(\tau))}{Q(\tilde{\theta}(\tau))} \right| < \varepsilon$, terminate the algorithm; or else

let $\tau^{\text{new}} = \tau + 1$ and continue step2-step3.

The algorithm 1 can be readily generalized to the case of multiple HRRP training samples, which can improve the parameter estimation accuracy.

In the recognition phase, it is required to calculate the likelihood $L(\tilde{\theta} | z')$ of the testing sample z' . By using Kalman forward filtering, the likelihood $L(\tilde{\theta} | z')$ can be computed as follows:

$$L(\tilde{\theta} | z') = \prod_{f=1}^d \frac{1}{(2\pi)^{1/2} (\tilde{\mathbf{C}}\Sigma_{f|f-1}\tilde{\mathbf{C}}^T + \tilde{\sigma}^2)^{1/2}} \exp\left[-\frac{(z_f' - \tilde{\mu} - \tilde{\mathbf{C}}\mathbf{u}_{f|f-1})^2}{2(\tilde{\mathbf{C}}\Sigma_{f|f-1}\tilde{\mathbf{C}}^T + \tilde{\sigma}^2)}\right] \quad (8)$$

where z_f' is the f th element of z' , $\mathbf{u}_f'$ is its corresponding hidden state, $\mathbf{u}_{f|f-1}' = E(\mathbf{u}_f' | \{z_1', z_2', \dots, z_{f-1}'\})$ and $\Sigma_{f|f-1}' = E[(\mathbf{u}_f' - \mathbf{u}_{f|f-1}')^T (\mathbf{u}_f' - \mathbf{u}_{f|f-1}') | \{z_1', z_2', \dots, z_{f-1}'\}]$ are the conditional mean and covariance matrix estimated by Kalman forward filtering.

4. Experimental Results

4.1 Data Description

The results presented in this paper are based on measured data of three planes. The center frequency and bandwidth of the radar are 5520MHz and 400MHz, respectively. The projections of plane trajectories onto ground plane are segmented as displayed in Figure 1. The fifth and sixth parts of An-26, the sixth and seventh parts of Cessna and the second and fifth parts of Yak-42 are chosen as training data with the remaining parts taken as testing data. There are 25000 HRRPs in each part except for the fifth part of Yak-42 which has 10000 HRRPs. In addition, the dimensionality of HRRP is 256.

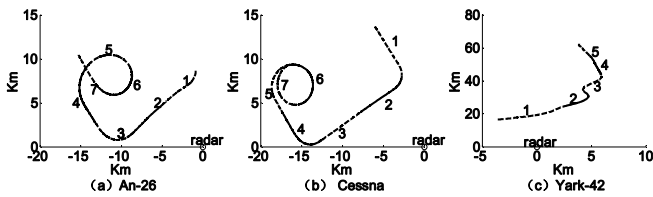


Fig.1. The projections of three target trajectories onto the ground plane.

To eliminate the target-aspect sensitivity [2], we adopt the equal interval partition method as in [2, 6] with frame length $N=1000$. Thus there are 50/50/35 frames for An-26/Cessna/Yark-42, respectively. Moreover, the spectral amplitudes of all HRRPs in each frame are $L2$ normalized [2] to remove the amplitude sensitivity.

4.2. The Effect of Hidden State Dimensionality to Recognition Performance

In this section, we employ LDM to model each frame of all targets and adopt algorithm 1 to estimate model parameters. The convergence threshold ε is set as 10^{-4} .

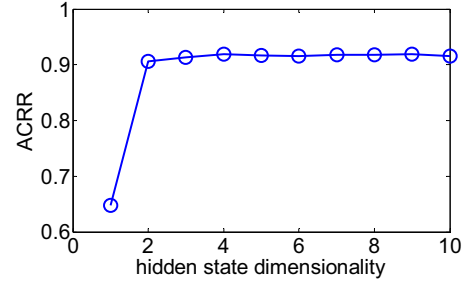


Fig.2. The ACRRs of LDM vs. hidden state dimensionality.

The average correct recognition rates (ACRR) against hidden state dimensionality is shown in Figure 2. As we can see when the hidden state dimensionality $m=1$, the ACRR is as low as 65.1%. This is due to the LDM here is too simple to capture the frequency evolution of spectral amplitude, i.e., the under-fitting problem [9]. With the increase of m , the LDM can describe the spectral amplitude more accurately and the ACRRs increase quickly. When $m=3$, the ACRR reaches 91.4%. Afterwards, the ACRRs tends to be stable. To balance the recognition accuracy and the computation load, we will fix $m=4$ in all the following experiments.

Figure 3 compares the recognition results of AGC model, FA model and LDM. The ACRR of LDM is better than the AGC model, while slightly inferior to the FA model. This establishes the viability of the proposed method.

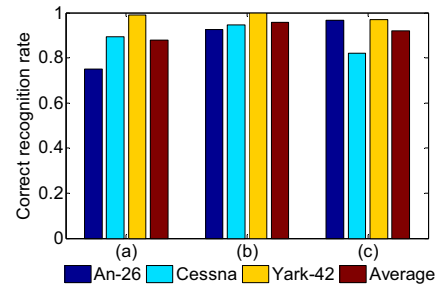
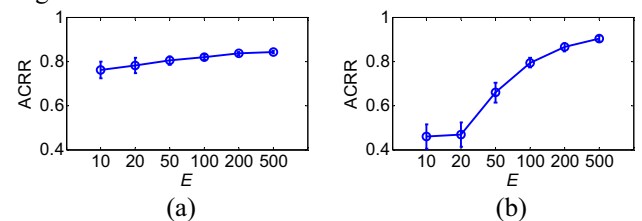


Fig 3. Correct recognition rates for three targets based on different models: (a) AGC model; (b) FA model; (c) LDM.

4.3. The Effect of Training Set Size to Recognition Performance

A practical RATR system should provide acceptable recognition accuracy even with a few training samples. To assess the recognition performance with respect to training set size, we carry out the following experiments: randomly pick E samples in each frame of all targets as training set and investigate the recognition accuracy of AGC model, FA model and LDM against different E values. Twenty trials are conducted for each E value and the results are displayed in Figure 4.



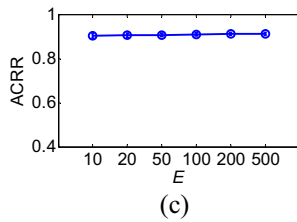


Fig. 4. The ACRRs of different models against training sample size: (a) AGC model; (b) FA model; (c) LDM.

From Figure 4 we can see the ACRR of the FA model deteriorates drastically with the decrease of training set size. When using a training set consisting of 1/50 of the total training samples, the FA model can hardly recognize targets. However, the ACRRs of LDM are quite robust to the variation of training set size. Even with a fraction of 1/100 of the total training samples (i.e., $E = 10$), the LDM can still successfully recognize the three targets. These results demonstrate that the LDM can greatly relax the training set size while still keep favorable recognition accuracy.

4.4 The Effect of Training Set Size to Rejection Performance

Rejecting samples not belonging to the training set classes (i.e., confusers) is another important index to evaluate the RATR system performance. In this experiment, 18000 HRRPs was generated by simulation software – XPATCH as confusers. To visually compare the rejection abilities of different models, we employ the receiver operating characteristics (ROC) curve in the signal detection theory.

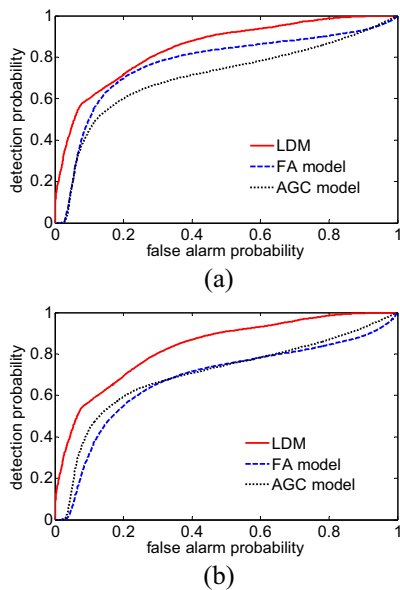


Fig. 5. ROC curves of three models based on different training set size: (a) $E=1000$ (b) $E=50$.

In Figure 5(a), we plot the ROC curves of three models using all the training samples. It is seen that the LDM always get higher detection probability than the others. To investigate the effect of training set size to the rejection performance, Figure 5(b) shows the ROC curves of three models with $E=50$. As we can see, the ROC curves of the AGC model and LDM are nearly the same, whereas that of the FA model changes evidently. The decrease in training set size worsens the

rejection ability of the FA model. This result further verifies that the proposed method can behave well under small training set size conditions.

5. Conclusion

A new radar target recognition method based on linear dynamic model was proposed to relax the limitation of training set size. Expectation Maximization algorithm was adopted to estimate model parameters. Experimental results based on measured data show the new method can achieve desirable recognition and rejection performance even with a very small training set size.

Acknowledgements

This work is partially supported by the National Natural Science Foundation of China (60901067 and 61001212), Program for New Century Excellent Talents in University (NCET-09-0630), Program for Changjiang Scholars and Innovative Research Team in University (IRT0954), and the Fundamental Research Funds for the Central Universities.

References

- [1] Jacobs S. P. and O'Sullivan J. A., "Automatic target recognition using sequences of high resolution radar range profiles," *IEEE Trans. Aerosp. Electron. Syst.*, 2000, 36(2): 364–380.
- [2] Du L., Liu H. and Bao Z., "Radar HRRP statistical recognition: parametric model and model selection," *IEEE Trans. Signal Process.*, 2008, 56(5): 1931–1944.
- [3] A. Zyweck and R. E. Bogner, "Radar target classification of commercial aircraft," *IEEE Trans. Aerosp. Electron. Syst.*, 1996, 32(2): 598–606.
- [4] X. J. Liao, P. Runkle, and L. Carin, "Identification of ground targets from sequential high-range-resolution radar signatures," *IEEE Trans. Aerosp. Electron. Syst.*, 2002, 38(4): 1230–1242.
- [5] Wong S. K., "Non-cooperative target recognition in the frequency domain," *IEE Proceedings – Radar, Sonar, and Navigation*, 2004, 151(2): 77–84.
- [6] Wang P., Dai F., Pan M., Du L. and Liu H., "Radar HRRP target recognition in frequency domain based on autoregressive model," *IEEE Radar Conference*, Kansas City, 2011: 714–717.
- [7] Shi L., Wang P., Liu H., Xu L. and Bao Z., "Radar HRRP Statistical Recognition With Local Factor Analysis by Automatic Bayesian Ying-Yang Harmony Learning," *IEEE Trans. Signal Process.*, 2011, 59(2): 610–617.
- [8] Wang P., Shi L., Du L., Liu H., Xu L. and Bao Z., "Radar HRRP statistical recognition with temporal factor analysis by automatic Bayesian Ying-Yang harmony learning," *Frontiers of Electrical and Electronic Engineering in China*, 2011, 6(2): 300–317.
- [9] R. O. Duda, P. E. Hart, and D. G. Stork. *Pattern classification*, John Willy and sons, Inc., New York, 2001.
- [10] H. L. Van Trees. *Detection, estimation, and modulation theory*, vol. 3, Wiley, New York, 1971.
- [11] Digalakis V., Rohlicek J., and Ostendorf M., "ML estimation of a stochastic linear system with the EM algorithm and its application to speech recognition," *IEEE Trans. Speech Audio Process.*, 1993, 1(4): 431–442.